

- ① Determine the built in barrier potential for silicon with the following data at room temperature.

$$N_A = 5 \times 10^{15} \text{ cm}^{-3}, \quad N_D = 2 \times 10^{15} \text{ cm}^{-3}$$

$$n_i = 1.5 \times 10^{10} \text{ cm}^{-3}, \quad V_{bi} = \left(\frac{KT}{q} \right) \ln \left(\frac{N_D N_A}{n_i^2} \right)$$

$$V_{bi} = 26 \times 10^{-3} \times \ln \left[\frac{5 \times 10^{21} \times 2 \times 10^{21}}{1.5 \times 1.5 \times 10^{32}} \right]$$

$$= 26 \times 10^{-3} \ln [4.44 \times 10^{10}]$$

$$= 26 \times 10^{-3} \times 24.507$$

$$= \underline{\underline{0.637 \text{ volt}}}$$

- ② The built in barrier potential of GaAs at room temperature is 1.08 volt. Determine the intrinsic carrier concentration if concentration of donors and acceptors are $2 \times 10^{15} \text{ cm}^{-3}$ and $5 \times 10^{15} \text{ cm}^{-3}$ respectively.

$$V_{bi} = \left(\frac{KT}{q} \right) \ln \left[\frac{N_D N_A}{n_i^2} \right]$$

$$\left[\frac{1.08}{26 \times 10^{-3}} \right] = \ln \left[\frac{N_D N_A}{n_i^2} \right]$$

$$e^{41.53} = \frac{N_D N_A}{n_i^2}$$

$$n_i^2 = \frac{2 \times 10^{21} \times 5 \times 10^{21}}{1.087 \times 10^{18}} = 9.199 \times 10^{24}$$

$$n_i = \underline{\underline{3.03 \times 10^{12} / \text{m}^3}}$$

(Dr. Santosh Mani)

③ A P^+n junction diode has acceptor concentration of $10^{20} / \text{cm}^3$ and intrinsic concentration of $10^{17} / \text{cm}^3$. Determine the built in barrier potential at room temperature if it is silicon.

for silicon $E_g = 1.12 \text{ eV}$

$$\begin{aligned} V_{bi} &= \frac{E_g}{2q} + \left(\frac{KT}{q} \right) \ln \left(\frac{N_D}{n_i} \right) \\ &= \frac{1.12 \times 1.6 \times 10^{-19}}{2 \times 1.6 \times 10^{-19}} + 26 \times 10^{-3} \ln \left(\frac{10^{26}}{10^{23}} \right) \\ &= 0.56 + 26 \times 10^{-3} \times \ln (10^3) \\ &= 0.56 + 0.1796 \end{aligned}$$

$$V_{bi} = 0.739 \text{ volt}$$

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- ④ A germanium diode has a reverse saturation current of $0.19 \mu A$. Find the current in the diode when it is forward biased with $0.3V$ at $27^\circ C$ ③
 $n=1$

$$I_0 = 0.19 \times 10^{-6} A, \quad V = 0.3 \text{ Volt},$$

$$V_T = \frac{KT}{q} = 26 \text{ mV} = 26 \times 10^{-3} V$$

$$I = I_0 \left[e^{\frac{V}{nV_T}} - 1 \right] \quad \frac{V}{V_T} = \frac{0.3}{26 \times 10^{-3}} = 11.53$$

$$I = 0.19 \times 10^{-6} \left[e^{11.53} - 1 \right]$$

$$I = 0.19 \times 10^{-6} [101722.11 - 1]$$

$$I = 19.32 \text{ mA}$$

- ⑤ The forward current in a silicon diode is 15 mA at $27^\circ C$. If reverse saturation current is 0.24 nA . What is forward bias voltage? $n=2$ for silicon diode

$$I = I_0 \left[e^{\frac{V}{nV_T}} - 1 \right]$$

$$\frac{I}{I_0} = \frac{15 \times 10^{-3}}{0.24 \times 10^{-9}} = 6.25 \times 10^7$$

$$e^{\frac{V}{nV_T}} = 62500001$$

$$\frac{V}{nV_T} = \ln () = 17.950$$

$$V = 933.43 \times 10^{-3} = 0.933 \text{ Volt}$$

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- ⑥ A germanium diode carries a current of 10 mA when it is forward biased with 0.2 V at 27°C . Find the ^(a) reverse saturation current (b) bias voltage required to get a current of 100 mA , $n=1$

$$I = 10 \times 10^{-3} \text{ A}, V = 0.2 \text{ Volt}$$

$$(a) \quad I = I_0 \left[e^{\frac{V}{nV_T}} - 1 \right]$$

$$\frac{V}{nV_T} = \frac{0.2}{1 \times 26 \times 10^{-3}} = 7.692$$

$$e^{\frac{V}{nV_T}} - 1 = 2189.75$$

$$I_0 = \frac{I}{\left[e^{\frac{V}{nV_T}} - 1 \right]} = \frac{10 \times 10^{-3}}{2189.75} = 4.56 \times 10^{-3} \times 10^{-3}$$

$$I_0 = 4.56 \times 10^{-6} \text{ A} = 4.56 \text{ } \mu\text{A}$$

$$(b) \quad V = ? \quad I = 100 \times 10^{-3} \text{ A}$$

$$I = I_0 \left[e^{\frac{V}{nV_T}} - 1 \right]$$

$$\frac{100 \times 10^{-3}}{4.56 \times 10^{-6}} = \left[e^{\frac{V}{nV_T}} - 1 \right]$$

$$\ln(21930.824) = \left(\frac{V}{nV_T} \right)$$

$$9.9956 = \frac{V}{1 \times 26 \times 10^{-3}}$$

$$V = 259.88 \times 10^{-3} = 0.259 \text{ Volt}$$

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⑦ A silicon diode has a saturation current of 1 pA at 22°C. What is bias voltage when diode current is 3.2 mA. What is bias current when temp changes to 102°C for same bias Voltage.

$$K = 1.38 \times 10^{-23} \text{ J/K}, n=2, I = 3.2 \times 10^{-3}$$

$$(i) I_0 = 1 \times 10^{-12} \text{ A}$$

$$V_T = \frac{KT}{q} = \frac{1.38 \times 10^{-23} \times 295}{1.6 \times 10^{-19}} = 0.02544 \text{ Volt}$$

$$I = I_0 \left[\left(e^{\frac{V}{nV_T}} \right) - 1 \right]$$

$$\ln \left(\frac{I}{I_0} + 1 \right) = \frac{V}{nV_T}$$

$$21.886 = \frac{V}{nV_T}$$

$$V = \underline{\underline{1.11 \text{ Volt}}}$$

$$(ii) V_T = \frac{1.38 \times 10^{-23} \times 375}{1.6 \times 10^{-19}} = 0.03234 \text{ Volt}$$

$$(I_0)_2 = (I_0), \quad 2^{\frac{(t_2 - t_1)}{10}} = 1 \times 10^{-12} \quad 2^{\frac{(102 - 22)}{10}} = 256 \times 10^{-12} \text{ A}$$

$$I_2 = (I_0)_2 \left[\left(e^{\frac{V}{nV_T}} \right) - 1 \right]$$

$$= 256 \times 10^{-12} \left[e^{17.16} - 1 \right]$$

$$I_2 = \underline{\underline{7.26 \text{ mA}}}$$

(6)

- ⑧ Find the dynamic resistance of P-N junction diode at forward current of 2mA. assume $\frac{KT}{q} = 25 \text{ mV}$

$$I = 2 \text{ mA} = 0.002 \text{ A}$$

$$V_T = 25 \times 10^{-3} \text{ V}$$

$$\eta = 1$$

$$r_d = \frac{n V_T}{I}$$

$$= \frac{1 \times 25 \times 10^{-3}}{2 \times 10^{-3}} = 12.5 \Omega$$

- ⑨ Determine the dc resistance for diode at

$$I_D = 2 \text{ mA},$$

$$I_D = 20 \text{ mA}$$

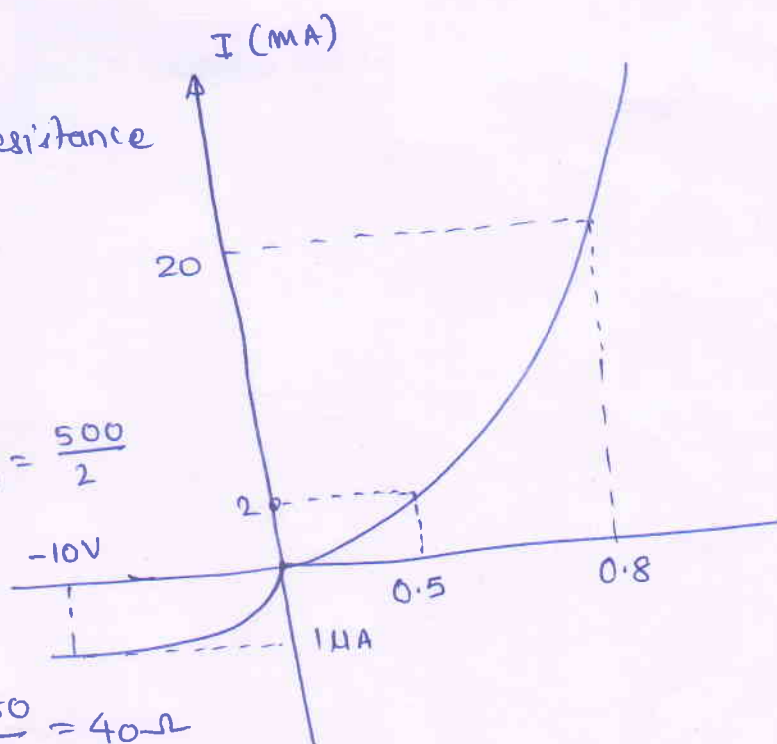
$$V_D = -10 \text{ V}$$

$$R_D = \frac{V_D}{I_D} = \frac{0.5}{2 \times 10^{-3}} = \frac{500}{2}$$

$$= 250 \Omega$$

$$R_D = \frac{0.8}{20 \times 10^{-3}} = \frac{80}{2} = 40 \Omega$$

$$R_D = \frac{10}{10^{-6}} = 10 \text{ M}\Omega$$



(Dr. Santosh Mani)